

Simon Marais Math Comp 2017 — A2

Jonathan Kasongo

October 24, 2025

Simon Marais Math Comp 2017 — A2

Let $a_1, a_2, a_3, \dots$ be the sequence of real numbers defined by $a_1 = 1$ and

$$a_m = \frac{1}{a_1^2 + a_2^2 + \dots + a_{m-1}^2} \quad \text{for } m \geq 2.$$

Determine whether there exists a positive integer N such that

$$a_1 + a_2 + \dots + a_N > 2017^{2017}.$$

Solution

Claim 1: The sequence (a_n) is always positive.

Proof: Since $a_1 = 1$ we have that $a_1^2 + a_2^2 + \dots + a_{m-1}^2 > 0$ and so by definition $a_m > 0$. $\square$

Claim 2: The sequence $(a_n)_{n=2}$ is decreasing.

Proof: We note that $a_1^2 + a_2^2 + \dots + a_{m-1}^2 = 1/a_m$. So

$$a_{m+1} = \frac{1}{\frac{1}{a_m} + a_m^2} = \frac{a_m}{1 + a_m^3}$$

and this is $< a_m$ iff $1/(1 + a_m^3) < 1$ and since $a_m > 0$ that inequality is always true. $\square$

Note that due to claim 1 we know that $\sum_{\text{sym}} a_1 a_2 > 0$ where that sum is symmetric over $(a_1, a_2, \dots, a_N)$. Now we can deduce that $(a_1 + a_2 + \dots + a_N)^2 > a_1^2 + a_2^2 + \dots + a_N^2 = 1/a_{N+1}$. Because of claim 2, we can make a_{N+1} be arbitrarily small. That is, let N be sufficiently large then we can make $1/a_{N+1} \rightarrow \infty$ and so we can make $(a_1 + a_2 + \dots + a_N)^2$ arbitrarily large.

That means that it is indeed possible to find a large positive integer N such that the inequality given in the problem statement is true. $\blacksquare$